

Model predictive control for nonlinear switched systems based on Koopman theory[☆]

Xuqian Ju^{a, }, Qing Sun^{a, *,1}, Dario Dennstädt^{b,c}, Karl Worthmann^b, Dajun Du^{a,1}, Minrui Fei^{a,1}

^a School of Mechatronic Engineering and Automation, Shanghai University, Shanghai, China

^b Institute of Mathematics, Optimization-based Control Group, Technische Universität Ilmenau, Ilmenau, Germany

^c Institut für Mathematik, Universität Paderborn, Paderborn, Germany

ARTICLE INFO

Keywords:

Model predictive control
Koopman operator
Nonlinear switched systems
Data-driven control design

ABSTRACT

This paper presents a novel data-driven predictive control strategy for discrete-time switched nonlinear systems. By leveraging the Koopman operator framework, we construct a linear surrogate model that effectively captures the systems' nonlinear dynamics, significantly enhancing prediction accuracy over standard Koopman methods that ignore switching behavior. We then formulate a Model Predictive Control (MPC) scheme based on this linear predictor. The proposed controller is designed for systems with known switching times, incorporating the sequence into the optimization horizon. Critically, we also formulate the controller to handle cases where the switching sequence is a decision variable, thereby enabling joint optimization of both the control input and the discrete mode switching. Numerical simulations demonstrate the method's feasibility and its significant advantages in both state prediction and control performance compared to traditional nonlinear and switched-system control strategies.

1. Introduction

Switched systems, which integrate multiple subsystems with dynamic transitions, are powerful modeling tools for complex real-world applications—from power grids [1] and aerospace vehicles [2] to biomedical devices [3] and mobile robots [4]. Their utility stems from the ability to switch between different dynamics based on predefined rules or system conditions. However, the mode-dependent nonlinearities and switching-induced time variation of these systems complicate fundamental tasks like accurate modeling, prediction, and controller design.

A major area of research in switched systems focuses on the inherent complexities of subsystem selection, switching-time determination, and input design. In many applications, the switching signal is not merely a disturbance but a key decision variable that can be actively chosen to enhance performance, manage constraints, or achieve tasks that are impossible under a single mode. Traditional control approaches such as backstepping [5], fault-tolerant control [6], Lyapunov-based

techniques [7], and neural-network-based control [8], have been extensively studied. Yet, these classical methods often struggle with computational complexity and uncertainty, particularly in systems with many modes, complex switching logic, or when control inputs and switching sequences must be co-designed. Consequently, they lack scalability for high-dimensional or highly nonlinear systems, limiting their practical applicability.

Data-driven methods, particularly the Koopman operator framework, offer a promising path forward. This approach transforms nonlinear systems into a higher-dimensional linear representation, facilitating analysis and control through linear system theory [9,10], see also the recent overview [11]. This lifting process relies on constructing a set of observables that form a Koopman-invariant subspace allowing the evolution of the nonlinear system to be represented through a linear operator in this lifted space, see e.g. [12–15] for an extension towards control systems.

The Koopman operator [16,17] and its numerical approximation [18], including the respective error analysis [19], have been extensively studied for single-mode dynamics, its application to switched systems

[☆] This work is supported by the National Natural Science Foundation of China (Grant No. 61803252, No. 62373240, No. 62273223). D. Dennstädt gratefully acknowledges funding by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project-ID 471539468.

* Corresponding author.

E-mail addresses: ju_xuqian@shu.edu.cn (X. Ju), qingsun@shu.edu.cn (Q. Sun), dario.dennstaedt@uni-paderborn.de (D. Dennstädt), karl.worthmann@tu-ilmenau.de (K. Worthmann), ddj@i.shu.edu.cn (D. Du), mrfei@staff.shu.edu.cn (M. Fei).

¹ Qing Sun, Dajun Du, and Minrui Fei are with the Shanghai Key Laboratory of Power Station Automation Technology, School of Mechatronic Engineering and Automation, Shanghai University, 200444, Shanghai, China.

<https://doi.org/10.1016/j.sysconle.2025.106340>

Received 26 March 2025; Received in revised form 1 December 2025; Accepted 28 December 2025

Available online 15 January 2026

0167-6911/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

remains relatively underexplored. The mode-dependent structure and switching transitions introduce unique complexities, such as the need to capture mode-dependent dynamics and handle mode transitions. Existing methods, such as [5,6,20], typically rely on either full knowledge of the switching sequences or restrictive assumptions about the system structure, limiting their practical applicability. In [21], switched-PDE models are reduced with mode-wise Koopman generators but rely on commuting dynamics. [22] combines multiple models with an automatic mode selector, while [23] employs neural observables to secure mean-square stability under stochastic switching. Other methods learn a latent-mode representation by switching between latent states [24]. These approaches either assume explicit mode selection [22], or focus on bilinear or affine structures [25] using generator blending by interpolation, or require large training sets and neural optimization [23,24]. Critically, a unified framework that efficiently co-designs control inputs and switching sequences while ensuring stability is still lacking. A similar comment applies to predictive control w.r.t. arbitrary, but known switching sequences.

To address these limitations, we propose a Koopman-based methodology tailored for switched nonlinear systems. Our approach constructs accurate, mode-specific linear predictors by incorporating known switching times using EDMDC [26]. These predictors are then integrated into a Model Predictive Control (MPC) framework [27], the switched Koopman-based MPC (S-KMPC), which handle both systems with exogenous switching signals and those where the switching sequence is a decision variable, enabling performance-driven mode switching. A key innovation is the guaranteed stability under arbitrary, including controller-optimized, switching via a common quadratic Lyapunov function (CQLF) in the lifted space. The principal contributions of this work are:

1. A switching-aware Koopman lifting procedure that uses a shared observable dictionary and a modified EDMDC to identify mode-consistent linear predictors. It achieves offset-free tracking for both predefined and controller-determined switching sequences, requiring no auxiliary mode selector and differing from neural or interpolated approaches [23,25].
2. A S-KMPC formulation endowed with a common quadratic Lyapunov function, guaranteeing mode-uniform recursive feasibility and closed-loop stability under arbitrary, including controller-optimized, switching sequences.
3. A computationally efficient pathway to robustness. Constructing a CQLF in the Koopman-lifted space is more tractable than for the original nonlinear switched system, providing a key enabler for stable control when the switching signal is actively chosen.
4. Numerical benchmarks on Van-der-Pol and DC-motor demonstrate higher prediction accuracy than classical single-mode Koopman MPC and nonlinear MPC baselines, while retaining real-time feasibility. In particular, we showcase the framework's ability to successfully solve control problems where optimizing the switching sequence is possible.

The paper is organized as follows. Section 2 introduces the problem formulation and details the design of Koopman-based predictors for switched systems. Section 3 describes the integration of these predictors into the MPC framework, including a proof of recursive feasibility. Section 4 presents numerical simulations to evaluate the proposed approach on benchmark switched systems. Finally, Section 5 concludes the paper and discusses potential avenues for future research.

Notation. $\mathbb{N}$ and $\mathbb{R}$ denote the sets of natural and real numbers, respectively, and we write $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. $\|x\| := \sqrt{\langle x, x \rangle}$ is the Euclidean norm of $x \in \mathbb{R}^n$. For a matrix $A = (a_{ij}) \in \mathbb{R}^{n \times m}$, $\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^m a_{ij}^2}$ denotes the Frobenius norm. If $A \in \mathbb{R}^{n \times n}$ is positive definite, then it is denoted as $A > 0$ and $\lambda_{\min}(A)$ refers to its smallest eigenvalue. $L^2(\mathbb{R}^m, \mathbb{R}^n)$ denotes the space of measurable square-integrable functions $f: \mathbb{R}^m \rightarrow \mathbb{R}^n$.

2. Koopman-based prediction

The prediction and control of nonlinear switched systems remains challenging due to mode-dependent nonlinear dynamics and mode transitions. We address these difficulties with a Koopman-operator framework that lifts the dynamics to a higher-dimensional using extended dynamic mode decomposition (EDMD) to generate a linear surrogate for each mode.

2.1. Recap on koopman and EDMD with control

The control of complex nonlinear discrete systems given by

$$x^+ = F(x, u) \quad (1)$$

with a nonlinear map $F: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ presents a significant challenge. Here, x and x^+ denote the current and the successor state, while u is the control value at the current time. Typically, the system is subject to state and control constraints, expressed as $x \in \mathbb{X} \subset \mathbb{R}^n$ and $u \in \mathbb{U} \subset \mathbb{R}^m$.

We make the following assumption about constraint sets.

Assumption 1 (Constraint Sets). The state and input constraint sets $\mathbb{X}$ and $\mathbb{U}$ are nonempty, compact polyhedra that contain the origin in their interiors:

$$\mathbb{X} = \{x \in \mathbb{R}^n \mid E_x x \leq b_x\}, \quad \mathbb{U} = \{u \in \mathbb{R}^m \mid E_u u \leq b_u\},$$

for $E_x \in \mathbb{R}^{p \times n}$, $E_u \in \mathbb{R}^{q \times m}$, $b_x \in \mathbb{R}^p$, and $b_u \in \mathbb{R}^q$ with $b_x > 0$ and $b_u > 0$ component-wise.

The Koopman operator yields a system representation through the lens of observable functions [28]. It gives an analytical way to view the nonlinear dynamical system as a linear one in a different, in general, infinite-dimensional space. We follow the approach proposed in [26,27], see also [29, Sections 1.4 and 1.5] for a concise summary. Given the input dependency in (1), we consider the Koopman operator on the space $\mathbb{R}^n \times \ell(\mathbb{U})$, where $\ell(\mathbb{U})$ is the space of all sequences $\mathbf{u} = (u_i)_{i=0}^\infty$. Then, the Koopman operator $\mathcal{K}$ is defined by

$$(\mathcal{K}\varphi)(x, \mathbf{u}) = \varphi(F(x, u(0)), \mathbf{u}(\cdot + 1)) \quad (2)$$

for all observables $\varphi \in L^2(\mathbb{R}^n \times \ell(\mathbb{U}), \mathbb{R})$, where $\mathbf{u}(\cdot + 1)$ denotes the shifted sequence of control values (corresponding to the left shift operator), see [27] for details. As pointed out in [29], the value $u = u(0)$ plays the role of a parameter and one may equivalently define a family $(\mathcal{K}_u)_{u \in \mathbb{R}^m}$ of Koopman operators on $\mathbb{R}^n$ parametrized by u , where each operator is associated with the map $F(\cdot, u)$, which corresponds to the Koopman identity $(\mathcal{K}_u \varphi)(x) = \varphi(F(x, u))$. To reduce the nonlinear control problem to a linear one, we apply the approach proposed in [27, Subsection 3.2.1] invoking linearity and causality, i.e., we choose functions $\psi_k \in L^2(\mathbb{R}^n, \mathbb{R})$, $M \in \mathbb{N}$, and combine them with control values $u_i \in \mathbb{U}$ resulting in

$$\{\psi_k(x)\}_{k=1}^M \times \{u_i\}_{i=1}^m. \quad (3)$$

Then, the line of arguments used in [27] shows that the lifted dynamics of the data-driven surrogate dynamics are linear, i.e.,

$$z^+ = Az + Bu \quad (4)$$

with $z = (\psi_1(x), \dots, \psi_M(x))^T$, $A \in \mathbb{R}^{M \times M}$, and $B \in \mathbb{R}^{M \times m}$. Usually, the functions ψ_k , $k \in [1 : M]$, can be regarded as *full state observable*, i.e., the set of observables (functions of the state) contains enough information to reconstruct the entire state of the system, see [18] for more details. We denote $\mathbb{V} := \text{span}(\{\psi_k : k \in [1 : M]\})$ as the L^2 -subspace of dimension M spanned by the dictionary of chosen observables.

Remark 1. For choice of observables and identification of A, B , we follow the guidelines in [18,27], employing a dictionary of one hundred thin-plate-spline radial basis functions

$$\psi_j(x) = \|x - c_j\|^2 \log(\|x - c_j\|), \quad j = 1, \dots, 100,$$

whose centres c_j are sampled once from the uniform distribution on $[-1, 1]^2$. With $z = \psi(x)$ we assemble the data blocks Ψ_X, Ψ_Y, U from snapshot triples (x_k, u_k, x_{k+1}) and solve the least-squares problem $\min_K \|\Psi_Y - K[\Psi_X; U]\|_F^2$. Partitioning the result as $K = [A \ B]$ yields the lifted matrices in (4), and the decoder C follows from (11).

The accuracy of many methods based on Koopman theory strongly depends on the choice of the dictionary, i.e., the observables that span the respective subspace $\mathbb{V}$, see [30]. More details on Koopman invariant spaces are given in [14] or the recent work [19], where Koopman invariance of the underlying reproducing kernel Hilbert space (RKHS) is rigorously shown if the observables are chosen as features corresponding to half-evaluated Wendland kernels. This results in the possibility of matrix representations on a finite-dimensional basis. The approximation of these matrices can then be conducted with a finite amount of data, which is used to construct a linear predictor for nonlinear systems.

We point out that the presented linear model corresponds to EDMDC (EDMD with control) and relies on the particular choice of observables (3), which may lead to limitations w.r.t. the approximation capability of the data-based surrogate model, see [31]. A potential remedy for (approximately) control-affine systems are bilinear surrogate models as proposed in [30], for which a rigorous analysis of the estimation and projection error is available in [32,33]. Here, we refer to [34,35] for (predictive) controller design methodologies. Moreover, we refer to [36] for an extension of the error analysis to reproducing kernel Hilbert spaces, which allows to establish uniform error bounds using Wendland kernels [37] and Koopman invariance [19].

2.2. Koopman and EDMDC for switched systems

In the following, we extend EDMDC (extended dynamic mode decomposition with control) to controlled *switched* systems. Based on a switching function, a formal system definition is the following, see [38, 39].

Definition 1 (Discrete-time Controlled Switched System). Let $\mathbb{X} \subseteq \mathbb{R}^n$ be the state space and $\mathbb{U} \subseteq \mathbb{R}^m$ the input set. Let $\mathcal{R} = \{1, \dots, m_r\}$ be a finite set of modes and, for each $r \in \mathcal{R}$, let $F_r : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{X}$ denote the mode-dependent dynamics. A *switching signal* is a map $\sigma : \mathbb{N}_0 \rightarrow \mathcal{R}$. The switched system evolves as

$$x_{k+1} = F_{\sigma(k)}(x_k, u_k), \quad x_0 \in \mathbb{X}, \quad u_k \in \mathbb{U}. \quad (5)$$

In this study, the switching times between modes are assumed to be either known a priori – based on heuristic rules derived from prior system knowledge – or decision variables for the controller. The use of heuristic rules for mode switching has been successfully applied in various control systems [40,41], which demonstrate the effectiveness of heuristic-based strategies in managing mode transitions without the complexity of real-time optimization. When the switching sequence is a decision variable, the optimization problem becomes even more complex, but with potentially significantly enhanced performance.

Remark 2 (Switching Signal and Stability Certificate). Treating the switching signal either as an exogenous signal or as a decision variable for the controller will allow, under **Assumption 5** (existence of a CQLF in lifted coordinates), a single quadratic function to certify stability for all modes and hence arbitrary switching.

In the ideal Koopman setting, there exists a finite-dimensional invariant subspace $\mathbb{V}$, which is made precise in the following.

Assumption 2 (Invariance of $\mathbb{V}$). Let the Koopman dictionary $\mathbb{V} = \text{span}(\{\psi_k \mid k \in [1 : M]\})$ be a finite-dimensional subspace such that $\mathcal{K}_{r,u}\mathbb{V} \subseteq \mathbb{V}$ for all modes $r \in \mathcal{R}$ and inputs $u \in \mathbb{U}$, leading to the mode-wise representation (4).

If **Assumption 2** holds, there exist, for each mode $r \in \mathcal{R}$, matrices $\mathcal{K}_r \in \mathbb{R}^{M \times M}$, $B_r \in \mathbb{R}^{M \times m}$, satisfying $\Psi(F_r(x, u)) = \mathcal{K}_r \Psi(x) + B_r u$ for all $(x, u) \in \mathbb{X} \times \mathbb{U}$, where Ψ is defined by $[\psi_1, \dots, \psi_M]^T$. **Assumption 2** is common when it comes to representing control systems with the Koopman operator; see, e.g. [42,43] and the recent work on exact Koopman liftings [44,45]. In this work, the Koopman operator for the discrete-time switched system is approximated separately for each mode $r \in \mathcal{R}$ to account for the mode-dependent dynamics, following a similar approach as for the nonlinear control system (1). Then, feedback-rectifiable stabilization of the switched linear system in the lifted space may be leveraged to ensure existence of a CQLF as, e.g., proposed in [46].

Next, we compose a switched EDMDC-based surrogate model.

Assumption 3. Let, for each mode $r \in \mathcal{R}$, d data points

$$(x_i, u_i, x_i^+, r) \in \mathbb{X} \times \mathbb{U} \times \mathbb{X} \times \mathcal{R}$$

satisfying $x_i^+ = F_r(x_i, u_i)$ be given.

For each mode $r \in \mathcal{R}$, using d data points collected at discrete time steps within the state constraint $x_1, \dots, x_d \in \mathbb{X} \subset \mathbb{R}^n$, we define the $(M \times d)$ matrices

$$X_{\text{lift}} := \begin{bmatrix} \psi_1(x_1) & \dots & \psi_1(x_d) \\ \vdots & & \vdots \\ \psi_M(x_1) & \dots & \psi_M(x_d) \end{bmatrix}, \quad (6)$$

$$Y_{\text{lift}}^r := \begin{bmatrix} (\mathcal{K}_r \psi_1)(x_1) & \dots & (\mathcal{K}_r \psi_1)(x_d) \\ \vdots & & \vdots \\ (\mathcal{K}_r \psi_M)(x_1) & \dots & (\mathcal{K}_r \psi_M)(x_d) \end{bmatrix}, \quad (7)$$

$$U := [u_1 \mid u_2 \mid \dots \mid u_d] \in \mathbb{R}^{m \times d}, \quad (8)$$

where U is a stacked matrix consisting of the control values u_i , $i \in [1 : d]$. The lifted observables ψ_k , $k \in [1 : M]$, are evaluated at the next state x^+ , which is obtained from the system dynamics. This captures the effect of the Koopman operator $\mathcal{K}_r$ on each observable ψ_k . Thus, Y_{lift}^r encodes the action of the Koopman operator $\mathcal{K}_r$ on the observables for each mode r , capturing state transitions. Importantly, Y_{lift}^r can be obtained from measurements of the system dynamics F_r using Eq. (5), without requiring explicit knowledge of the operator $\mathcal{K}_r$. This highlights that the transformation leverages observable evaluations rather than direct access to the Koopman operator itself. Therefore, a linear predictor for nonlinear switched systems can be constructed as follows

$$\begin{cases} Y_{\text{lift}}^r = \mathcal{K}_r X_{\text{lift}} + B_r U \\ \hat{x} = C X_{\text{lift}}, \end{cases} \quad (9)$$

where $\mathcal{K}_r, B_r$ are obtained by

$$[\mathcal{K}_r \mid B_r] := \arg \min_{\mathcal{K}_r \in \mathbb{R}^{M \times M}, B_r \in \mathbb{R}^{M \times m}} \|\mathcal{K}_r X_{\text{lift}} + B_r U - Y_{\text{lift}}^r\|_F^2. \quad (10)$$

To compute the state prediction $\hat{x} \in \mathbb{R}^n$, let $X = [x_1 \mid x_2 \mid \dots \mid x_d] \in \mathbb{R}^{n \times d}$, be the matrix containing the d state samples. Then, C is obtained via the optimization of

$$C := \arg \min_{C \in \mathbb{R}^{n \times M}} \|C X_{\text{lift}} - X\|_F^2. \quad (11)$$

Using the lifted predictor (9), the least-squares fit (10) provides mode-specific matrices $\mathcal{K}_r, B_r$, while (11) yields a common decoder C that reconstructs the original state from the lifted one. Because the Koopman operator acts on observables rather than states, we approximate it in a finite dictionary via EDMD [18,47]. Doing so per mode captures the abrupt changes in switched dynamics yet keeps every subsystem linear in the lifted space. The resulting compact predictors – refined online if desired – enable accurate, real-time modeling and control of switched nonlinear systems.

Remark 3. Our Koopman predictors are identified offline from the data available for each mode, compare [Assumption 3](#). If the system is later operated with additional snapshots continually appearing, it is possible to refine each mode's predictor on the fly. The dictionary of basis functions is kept unchanged; only mode's Koopman matrix is updated through a simple recursive least-squares step that assimilates the newest snapshot pair and gradually down-weights old data. This incremental adjustment runs in $\mathcal{O}(M^2)$ time per update – well within one sampling period – and has been shown to restore accuracy as fresh data accumulate, see [\[48,49\]](#). Also, note that interpolation between Koopman generators [\[25\]](#) can smooth control in affine settings, but is not required in the present framework because each nonlinear mode is modeled explicitly.

2.3. Choice of basis functions

In EDMD, the nonlinear state is embedded through a dictionary of observables, and the chosen functions largely determine both fidelity and cost. For switched nonlinear systems, one typically mixes low-order polynomials, Gaussian-type radial functions for strongly varying regions, Fourier modes to capture periodic content, and – when data are plentiful – delay coordinates or neural features learned from trajectories. The specific ones depend on the exact characteristics of the subsystems.

There is an inherent trade-off: adding more basis functions usually improves the model's expressive power, yet each extra term increases the size of the lifted state, the regression problem, and the online evaluation cost. In practice one expands the dictionary only until further additions yield negligible reduction in prediction error on held-out data, or until an information criterion (e.g. Akaike information criterion [\[50\]](#)) starts rising, then verifies that the resulting dimension still meets the real-time budget of the target hardware. This simple “marginal-gain versus cost” rule keeps the model accurate enough for control while preventing unnecessary computational load.

If the chosen dictionary does not span a Koopman-invariant subspace (i.e., [Assumption 2](#) fails), prediction accuracy deteriorates; kernel-based extensions such as [\[14,51\]](#) mitigate this issue but still rely on selecting a suitable dictionary—an open challenge when no prior knowledge of the nonlinear dynamics is available.

This paper proposes a method for learning observables $\{\psi_1, \dots, \psi_M\}$ that span an invariant subspace of the Koopman operator, based on a dataset of measurement sequences, inspired by the work of Naoya Takeishi et al. [\[12\]](#).

The Koopman approximation consists of lifting Ψ , which generates the subspace $\mathbb{V}$, and the compressions $P_{\mathbb{V}} \mathcal{K}_r|_{\mathbb{V}}$, $r \in \mathcal{R}$, is computed based on the RSS loss $\mathcal{L}_{\text{RSS}} : L^2(\mathbb{R}^n, \mathbb{R}^M) \times \mathbb{R}^{m \times d} \times \mathbb{R}^{m \times d} \times \mathbb{N} \rightarrow \mathbb{R}$

$$\mathcal{L}_{\text{RSS}}(\Psi; X, U, M) = \sum_{j=1}^d \|x_j - C\Psi(x_j)\|^2 + \lambda \sum_{r=1}^{m_r} \sum_{j=1}^d \|\mathcal{K}_r \Psi(x_j) + B_r u_j - \Psi(F_r(x_j, u_j))\|^2, \quad (12)$$

where $\Psi : \mathbb{R}^n \rightarrow \mathbb{R}^M$, $\Psi(x) = (\psi_1(x), \dots, \psi_M(x))^T$ denotes the observables (basis functions) learned by a multilayer perceptron (MLP) network. We emphasize the relation between the dictionary Ψ and the matrices C , $\mathcal{K}_r$, B_r , $r \in \mathcal{R}$. If Ψ changes, $\mathcal{K}_r$, B_r , and C are updated according to the regression problems [\(10\)](#) and [\(11\)](#). These observables approximate the span of a Koopman invariant subspace, i.e., a functional space that remains invariant under the action of the Koopman operator $\mathcal{K}_r$. A key difficulty is the simultaneous (approximate) invariance of $\mathbb{V}$ w.r.t. all approximants $\mathcal{K}_r$. Here, $F_r(x_j, u_j)$ represents the state at the next time step under the switched system dynamics, and $\lambda > 0$ is a regularization parameter enforcing invariance.

The MLP network is trained to minimize $\mathcal{L}_{\text{RSS}}$, which jointly optimizes:

1. State Reconstruction: The term $\|x_j - C\Psi(x_j)\|^2$ ensures the observables Ψ span a subspace where the original state x can be linearly reconstructed via C .
2. Invariance Enforcement: The term $\|\mathcal{K}_r \Psi(x_j) + B_r u_j - \Psi(F_r(x_j))\|^2$ penalizes deviations from the Koopman operator's linearity, ensuring the subspace remains invariant.

Remark 4. When the state x is not measured directly, we observe $y = \phi(x)$. A standard delay embedding

$$\tilde{x}_k = [y_k^T, y_{k-\tau}^T, \dots, y_{k-(d_e-1)\tau}^T]^T$$

yields a diffeomorphic copy of the true state whenever $d_e > 2 \dim x$ [\[52, 53\]](#). Thus, $\tilde{x}_k$ can be used as the input to the Koopman-learning pipeline; [Fig. 1](#) sketches the resulting data flow.

2.4. Choosing basis functions via a neural network

Section [2.3](#) formulated the residual-sum-of-squares loss [\(12\)](#), which couples state reconstruction with Koopman invariance. In practice, we realize Ψ (and, if needed, the reconstructor h) with compact multilayer perceptrons (MLPs [\[54\]](#)) that use PReLU activations [\[55\]](#). Hyperparameters – delay length, embedding size, observable dimension – are tuned once per application, keeping M small enough for real-time MPC.

Remark 5. A feed-forward MLP maps the (optionally delay-embedded) measurement to the observable vector $\Psi(x) \in \mathbb{R}^M$. Its **input** is the current embedded output, its **output** the M -dimensional observables, and its only **parameters** are shared weights and biases learned offline by minimizing the RSS loss [\(12\)](#).

The networks are trained offline with first-order stochastic gradient descent and batch normalization. Although global convergence cannot be proven, empirical evidence shows that the RSS loss reaches a satisfactory minimum after a few hundred epochs, producing Koopman matrices whose prediction accuracy matches EDMD while allowing fully data-driven feature learning. Reinforcement-learning refinements for switched Koopman models (see [\[56\]](#)) are a potential topic for future work.

3. Koopman-based model predictive control

In this section, we propose a switched Koopman-based MPC (S-KMPC) controller for the nonlinear switched system [\(5\)](#) using the linear predictor [\(9\)](#).

In MPC, at each time instant, the control input is determined as a solution of an optimization problem on a finite prediction horizon depending on the most-recent (state) measurement. Then, the procedure is repeated ad infinitum. In linear-quadratic MPC, a quadratic program is solved, thereby enabling rapid calculation of the resulting control input. Nonlinear Model Predictive Control (NMPC, see [\[57\]](#)) conversely, addresses a challenging non-convex optimization problem and therefore requires significant computational resources.

Just like linear MPC, Koopman MPC [\[27\]](#) and the proposed extension to switched systems are based on quadratic programming.

3.1. Switched koopman-based MPC

Denote the lifted state as $x_{\text{lift}} = \Psi(x)$.

Assumption 4 (Lifting and Reconstruction). The lifting map $\Psi : \mathbb{X} \rightarrow \mathbb{R}^M$ is locally Lipschitz, its components $\{\psi_i\}_{i=1}^M$ are linearly independent in L^2 , and $\Psi(0) = 0$. Moreover, there exists a linear reconstruction map $C \in \mathbb{R}^{n \times M}$ such that

$$x = C \Psi(x) \quad \forall x \in \mathbb{X}.$$

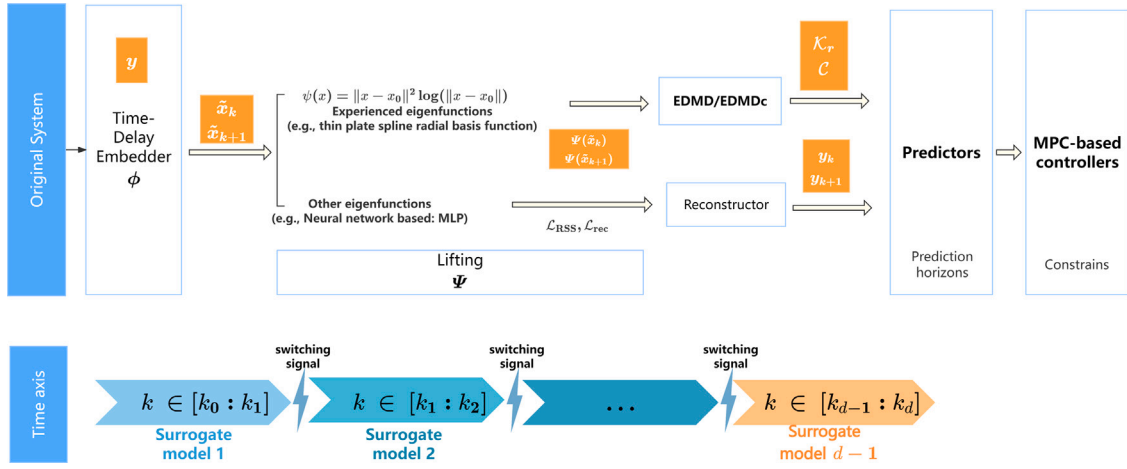


Fig. 1. Overview of the proposed framework. y represents the measurement of the original system, while $\tilde{x}_k, \tilde{x}_{k+1}$ denote the state representations. Ψ and h are the lifting functions and the reconstructor, respectively. The variable k represents time, with $k_0, k_1, \dots, k_d$ denoting specific time instants. The loss functions $\mathcal{L}_{RSS}$ and $\mathcal{L}_{rec}$ are optimized using a deep learning approach. Surrogate models are constructed based on the learned predictors.

Under [Assumption 4](#), the state constraints in the original coordinates are equivalently written as

$$E_x C x_{\text{lift}} \leq b_x,$$

while the input constraints remain unchanged, i.e., $E_u u \leq b_u$. With the above [Assumptions 3](#) and [4](#) and the lifted linear systems (9), $r \in [1 : m_r]$, the proposed switched Koopman-based MPC solves the following optimization problem at every time instant provided the switching sequence $\hat{\sigma}$ is a decision variable of the controller:

$$\begin{aligned} & \text{minimize}_{u_i, x_{\text{lift}}(i), \hat{\sigma}_i} J \left((u_i)_{i=k}^{k+N_h-1}, (x_{\text{lift}}(i))_{i=k}^{k+N_h} \right) \\ & \text{subject to} \quad x_{\text{lift}}(k) = \Psi(x_k), x_{\text{lift}}(k+N_h) \in \hat{\mathbb{X}}_f, \text{ and,} \\ & \quad \text{for } i \in [k : N_h - 1] : u_i \in \mathbb{U}, \hat{\sigma}_i \in \mathcal{R}, \\ & \quad x_{\text{lift}}(i+1) = \mathcal{K}_{\hat{\sigma}_i} x_{\text{lift}}(i) + B_{\hat{\sigma}_i} u_i, \\ & \quad E_x C x_{\text{lift}}(i+1) \leq b_x. \end{aligned} \quad (13)$$

where $N_h \in \mathbb{N}_0$ is the prediction horizon. If the decision variable cannot be modified but is a priori known instead, then the constraint $\hat{\sigma}_i = \sigma(i)$ for all $i \in [k : N_h - 1]$ is added as an additional constraint to the optimization problem. The terminal region $\hat{\mathbb{X}}_f$ in the lifted space is defined as

$$\hat{\mathbb{X}}_f := \{x_{\text{lift}} \in \mathbb{R}^M \mid V_f(x_{\text{lift}}) \leq \epsilon\}, \quad (14)$$

where $V_f(x_{\text{lift}}) = x_{\text{lift}}^\top P x_{\text{lift}}$ is often a common quadratic Lyapunov function (CQLF). Since the lifted dynamics are linear, we adopt the common quadratic form with $P = P^\top > 0$, and $\epsilon > 0$ is a constant defining the size of the terminal set; any (possibly non-quadratic) control Lyapunov function that renders the terminal set positively invariant could be used instead, but the quadratic choice yields a simple, mode-uniform synthesis for S-KMPC.

The quadratic cost function J is given by

$$\begin{aligned} & J \left((u_i)_{i=k}^{k+N_h-1}, (x_{\text{lift}}(i))_{i=k}^{k+N_h} \right) \\ & = V_f(x_{\text{lift}}(k+N_h)) + \sum_{i=k}^{k+N_h-1} (x_{\text{lift}}^\top(i) Q x_{\text{lift}}(i) + u_i^\top R u_i). \end{aligned} \quad (15)$$

The positive-definite matrices $Q \in \mathbb{R}^{M \times M}$ and $R \in \mathbb{R}^{m \times m}$ weight the state and the control input in the cost function, respectively. The matrices $E_x \in \mathbb{R}^{\tilde{n} \times n}$ and $E_u \in \mathbb{R}^{\tilde{m} \times m}$, along with the vectors $b_x \in \mathbb{R}^{\tilde{n}}$ and $b_u \in \mathbb{R}^{\tilde{m}}$, represent the polyhedral constraints on the state and input, $\tilde{n}, \tilde{m} \in \mathbb{N}$. In the following, we will use the shorthand notation $\hat{\mathbb{X}}$ to refer to the set constrained region in the lifted space, i.e.

$$\hat{\mathbb{X}} := \{x_{\text{lift}} \in \mathbb{R}^M \mid E_x C x_{\text{lift}} \leq b_x\}. \quad (16)$$

The optimization problem (13) implicitly defines a nonlinear feed-back controller $\kappa : \mathbb{R}^n \times \mathbb{N}_0 \rightarrow \mathbb{R}^m$

$$\kappa(x_k, k) = u_0^*(x_k, k), \quad (17)$$

and a switching sequence $\hat{\sigma} : \mathbb{R}^n \times \mathbb{N}_0 \rightarrow \mathcal{R}$, with $\hat{\sigma}(x_k, k) = \sigma_0^*(x_k, k)$ where $u_0^*(x_k, k)$ and $\sigma_0^*(x_k, k)$ denote an optimal solution to problem (13) parametrized by the current state x_k and the time instant k . In contrast, in the terminal region we employ the linear feedback in lifted coordinates

$$u = K_r^f x_{\text{lift}},$$

which underlies the construction of $A_r = \mathcal{K}_r + B_r K_r^f$, the CQLF V_f , and the terminal set $\hat{\mathbb{X}}_f$. The closed-loop operation of the lifting-based MPC is described in [Algorithm 1](#), where u^* represents the optimal control input and σ^* the optimal switching sequence.

Algorithm 1 S-KMPC

Input: Constraint sets $\mathbb{X}, \mathbb{U}$ as in [Assumption 1](#); a lifting map Ψ and reconstruction matrix C satisfying [Assumption 4](#); and data (x_i, u_i, x_i^+, r) , $i \in [1 : d]$, for each mode $r \in \mathcal{R}$ with $x_i^+ = F_r(x_i, u_i)$, satisfying [Assumption 3](#).

- 1: Compute $\mathcal{K}_r$, B_r , and C with (10) and (11).
- 2: At each time step $k \in \mathbb{N}_0$: Measure the state x_k and set the lifted state $x_{\text{lift}}(k) = \Psi(x_k)$, where Ψ is the lifting function using a set of observables $\{\psi_j\}_{j=1}^M$.
- 3: Solve (13) with (15) to obtain an optimal sequence of input values $(u_i^*)_{i=k}^{k+N_h-1}$ and associated switching sequence $(\sigma_i^*)_{i=k}^{k+N_h-1}$. Set $u_k = u_0^*(x_k, k)$ and apply it to the switched system (5) with $\hat{\sigma}(k) = \sigma_0^*$.

3.2. Recursive feasibility

We recall that the nominal lifted model associated with the predictor (9) is

$$\begin{cases} x_{\text{lift}}(k+1) = \mathcal{K}_{\hat{\sigma}(k)} x_{\text{lift}}(k) + B_{\hat{\sigma}(k)} u_k, \\ \hat{x}_k = C x_{\text{lift}}(k), \end{cases} \quad (18)$$

with the initial condition

$$x_{\text{lift}}(k) = \Psi(x_k), \quad (19)$$

where x_k denotes the state of the original system at time $k \in \mathbb{N}$ and Ψ is the lifting map. Here, $\hat{\sigma}(k)$ denotes the mode at time k chosen by the controller if the switching sequence is a decision variable and the

a priori known $\sigma(k)$ otherwise. The variable $x_{\text{lift}}(k)$ is the lifted state propagated by (18), u_k is the input, and C maps x_{lift} to the predicted state $\hat{x}_k$. Throughout this section, we fix the lifted predictor $(K_r, B_r)_{r \in \mathcal{R}}$ constructed from data under Assumptions 3 and 4. The lifted feasible sets $\hat{\mathbb{X}}$ and $\hat{\mathbb{X}}_f$ are induced by Assumptions 1 and 4.

To make the subsequent analysis precise, we impose the following standing assumption on the closed-loop lifted modes

$$x_{\text{lift}}(k+1) = (K_r + B_r K_r^f) x_{\text{lift}}(k), \quad (20)$$

Assumption 5. There exist feedback matrices $K_r^f \in \mathbb{R}^{m \times M}$, $r \in \mathcal{R}$, and a symmetric positive definite matrix $P \in \mathbb{R}^{M \times M}$ such that, with $A_r := K_r + B_r K_r^f$,

$$A_r^T P A_r - P < 0 \quad \text{for all } r \in \mathcal{R}.$$

Equivalently, $V_f(x_{\text{lift}}) = x_{\text{lift}}^T P x_{\text{lift}}$ is a common quadratic Lyapunov function (CQLF) for the closed-loop modes (20).

For the rest of the section, we fix feedback matrices $K_r^f \in \mathbb{R}^{m \times M}$, $r \in \mathcal{R}$, and a matrix $P > 0$ as in Assumption 5. We use these to define the quadratic terminal cost V_f and the terminal set $\hat{\mathbb{X}}_f$ as in (14). To analyze the switched system (18), we employ the common quadratic Lyapunov function V_f for all closed-loop modes (20).

Assumption 5 is strong (sufficient but not necessary). Existence of CQLFs has been studied; see, e.g., [58,59], see also the recent work [46] and the references therein. When Assumption 5 does not hold, less conservative guarantees under constrained switching are available via multiple-Lyapunov or average-dwell-time certificates [60,61].

Remark 6. In the original nonlinear coordinates, constructing a common quadratic Lyapunov function for multiple modes is generally non-convex. In the lifted coordinates, each mode shares the same quadratic form $V(x_{\text{lift}}) = x_{\text{lift}}^T P x_{\text{lift}}$ obtained from the LMIs $(A_r^T P A_r - P) < 0$. The availability of such a single quadratic Lyapunov function for all modes is a key benefit of working in Koopman space.

Given a common quadratic Lyapunov function, the set $\hat{\mathbb{X}}_f$ in (14) can be chosen positively invariant w.r.t. the dynamics (20) under state and input constraints, for a suitable $\epsilon > 0$, i.e.,

$$A_r x_{\text{lift}} \in \hat{\mathbb{X}}_f \quad \text{and} \quad K_r^f x_{\text{lift}} \in \mathbb{U}$$

for all $x_{\text{lift}} \in \hat{\mathbb{X}}_f \subseteq \hat{\mathbb{X}}$ and all $r \in \mathcal{R}$. Thus, for any state-mode pair (x_{lift}, r) with $x_{\text{lift}} \in \hat{\mathbb{X}}_f$, the feedback $u = K_r^f x_{\text{lift}} \in \mathbb{U}$ keeps the successor state $A_r x_{\text{lift}}$ in $\hat{\mathbb{X}}_f$. This invariance property is key for establishing recursive feasibility.

In the following recursive feasibility and stability analysis, we work in the ideal Koopman setting of Assumption 2, i.e., the finite-dimensional lifted predictor $(K_r, B_r, C)_{r \in \mathcal{R}}$ exactly represents the switched nonlinear dynamics (5) on $\hat{\mathbb{X}}$. Consequently, we treat the lifted model (18) as the plant dynamics.

Theorem 1 (Recursive Feasibility of S-KMPC).

Let Assumption 2 hold. Further, assume that Assumptions 3 and 4 hold with $d \in \mathbb{N}$ sufficiently large such that the exact Koopman lifting of Assumption 2 in Representation (4) is realized in Algorithm 1. The, if in addition Assumption 5 holds and the constant $\epsilon > 0$ is chosen sufficiently small such that, for each state-mode pair $(x_{\text{lift}}, r) \in \hat{\mathbb{X}}_f \times \mathcal{R}$, $\hat{\mathbb{X}}_f \subseteq \hat{\mathbb{X}}$ defined by (14), the inclusions

$$\bullet \quad K_r^f x_{\text{lift}} \in \mathbb{U} \quad \text{and} \quad A_r x_{\text{lift}} \in \hat{\mathbb{X}}_f \quad \text{with} \quad A_r = K_r + B_r K_r^f$$

hold. Then, the optimization problem (13) is recursively feasible. In particular, if optimization problem (13) is initially feasible, i.e., feasible at time $k = 0$, the optimization problem (13) remains feasible at all subsequent time instants $k \in \mathbb{N}$.

Proof. Assumption 5 implies the existence of a CQLF $V_f(x_{\text{lift}}) = x_{\text{lift}}^T P x_{\text{lift}}$, where $P \in \mathbb{R}^{M \times M}$ is symmetric and positive definite ($P > 0$). Suppose that an optimal solution $(u^*, x_{\text{lift}}^*, \sigma^*)$ to the optimization problem (13) and (15) exists at time $k \in \mathbb{N}_0$. This optimal solution has the following properties.

- The state and control constraints are satisfied:

$$x_{\text{lift}}^*(k+i) \in \hat{\mathbb{X}}, \quad u_{k+i}^* \in \mathbb{U}, \quad \forall i = 1, \dots, N_h - 1.$$

- The terminal constraint holds:

$$x_{\text{lift}}^*(k+N_h) \in \hat{\mathbb{X}}_f.$$

Define the control sequence $(u_i)_{i=k+1}^{k+N_h}$ and the associated switching sequence $(\hat{\sigma}_i)_{i=k+1}^{k+N_h}$ by $u_i = u_{i-1}^*$, $\hat{\sigma}_i = \sigma_{i-1}^*$ for $i = k+1, \dots, k+N_h-1$ and $u_{k+N_h} = K_{\hat{\sigma}(k+N_h)}^f x_{\text{lift}}^*(k+N_h)$, $\hat{\sigma}_{k+N_h} = \sigma(k+N_h)$. We show that, after applying u^* and σ^* to the switched system (18) for one time instance, $(u_i)_{i=k+1}^{k+N_h}$ and $(\hat{\sigma}_i)_{i=k+1}^{k+N_h}$ are a feasible control and switching sequence at the next time $k+1$. When applying the control $(u_i)_{i=k+1}^{k+N_h}$ and switching sequence $(\hat{\sigma}_i)_{i=k+1}^{k+N_h}$ to the system (18), we have

$$x_{\text{lift}}(k+1+i) = x_{\text{lift}}^*(k+1+i)$$

for all $i = 1, \dots, N_h - 1$. In particular, this implies $x_{\text{lift}}(k+N_h) \in \hat{\mathbb{X}}_f$. As $\bigcup_{r=1}^{m_r} \{K_r^f \hat{\mathbb{X}}_f\} \subset \mathbb{U}$ by assumption, we therefore have $K_{\hat{\sigma}(k+N_h)}^f x_{\text{lift}}(k+N_h) \in \mathbb{U}$. Thus, $u_{k+1+i} \in \mathbb{U}$ for all $i = 1, \dots, N_h - 1$. The decrease condition of the common quadratic Lyapunov function V_f yields

$$\begin{aligned} V_f(x_{\text{lift}}(k+1+N_h)) - V_f(x_{\text{lift}}(k+N_h)) \\ = V_f(x_{\text{lift}}(k+1+N_h)) - x_{\text{lift}}^T(k+N_h) P x_{\text{lift}}(k+N_h) < 0, \end{aligned}$$

since $x_{\text{lift}}(k+N_h) \in \hat{\mathbb{X}}_f$ and $x_{\text{lift}}(k+1+N_h) = (K_{\hat{\sigma}(k+N_h)} + B_{\hat{\sigma}(k+N_h)} K_{\hat{\sigma}(k+N_h)}^f) x_{\text{lift}}(k+N_h)$, which immediately gives $V_f(x_{\text{lift}}(k+1+N_h)) \leq \epsilon$. Thus, $x_{\text{lift}}(k+1+N_h) \in \hat{\mathbb{X}}_f \subset \hat{\mathbb{X}}$. Therefore, the control sequence $(u_i)_{i=k+1}^{k+N_h}$ and the associated switching sequence $(\hat{\sigma}_i)_{i=k+1}^{k+N_h}$ are feasible for the optimization problem (13) at time $k+1$, which proves recursive feasibility. $\square$

Theorem 2 (Stability). Let the assumptions of Theorem 1 hold. If the Common Quadratic Lyapunov Function $V_f(x_{\text{lift}}) = x_{\text{lift}}^T P x_{\text{lift}}$ with symmetric positive definite matrix $P \in \mathbb{R}^{M \times M}$ fulfills

$$V_f(A_r x) - V_f(x) + x^T Q x + (K_r^f)^T R (K_r^f x) < 0$$

for all $x \in \hat{\mathbb{X}}_f \setminus \{0\}$ and all modes $r \in \mathcal{R}$, then system (18) is asymptotically stable under the control law of the S-KMPC Algorithm 1.

Proof. Firstly, note that the S-KMPC Algorithm 1 is recursively feasible according to Theorem 1. Therefore, the optimization problem (13) has a solution at every time instance $k \in \mathbb{N}$ during the application of the S-KMPC algorithm to the system (18).

The solution of problem (13) at time $k \in \mathbb{N}$, i.e. an optimal input-state sequence with associated switching sequence, is denoted by $(u^*, x_{\text{lift}}^*, \sigma^*)$ with optimal value $J(u^*, x_{\text{lift}}^*) \geq 0$, cp. (15). We show J exhibits a Lyapunov decrease along the MPC closed-loop trajectory.

Let $k \in \mathbb{N}$ be arbitrary but fixed. Note that the control sequence $(\tilde{u}_i)_{i=k+1}^{k+N_h}$ with associated switching $(\tilde{\sigma}_i)_{i=k+1}^{k+N_h}$ given by $\tilde{u}_i = u_{i-1}^*$ and $\tilde{\sigma}_i = \sigma_{i-1}^*$ for $i = k+1, \dots, k+N_h-1$ and $u_{k+N_h} = K_{\tilde{\sigma}(k+N_h)}^f x_{\text{lift}}^*(k+N_h)$, $\tilde{\sigma}_{k+N_h} = \sigma(k+N_h)$ is a feasible control at time $k+1$. We denote the associated state sequence by $\tilde{x}_{\text{lift}}$ and point out that $x_{\text{lift}}^*(i+1) = \tilde{x}_{\text{lift}}(i)$ holds for $i = k, \dots, k+N_h-1$. Hence, we get

$$\begin{aligned} J^*(k+1) - J^*(k) &\leq J(\tilde{u}, \tilde{x}_{\text{lift}}) - J^*(u^*, x_{\text{lift}}^*) \\ &= V_f(A_r \tilde{x}_{\text{lift}}) - V_f(\tilde{x}_{\text{lift}}) + \tilde{x}_{\text{lift}}^T (Q + (K_r^f)^T R K_r^f) \tilde{x}_{\text{lift}} \\ &\quad - (x_{\text{lift}}^*(k)^T Q x_{\text{lift}}^*(k) + \|u_k^*\|_R^2) \\ &\leq -x_{\text{lift}}^*(k)^T Q x_{\text{lift}}^*(k) \leq -\lambda_{\min}(Q) \|x_{\text{lift}}^*(k)\|^2 \end{aligned}$$

using the notations $\bar{x}_{\text{lift}} = x_{\text{lift}}^*(k + N_h) = \bar{x}_{\text{lift}}(k + N_h - 1)$ and $\bar{r} = \bar{\sigma}_{k+N_h} = \sigma(k + N_h)$. That shows the desired Lyapunov decrease and, thus, asymptotic stability.² In particular, the closed-loop trajectory converges to the desired steady state. $\square$

A key ingredient of the presented stability proof is that J does not explicitly depend on time. This is only implicitly taken into account by solving the optimization problem (13). Due to the assumed CQLV and its uniformity w.r.t. the mode sequence, we were able to leverage the usual arguments to conclude asymptotic stability of the MPC closed loop.

Remark 7. The requirement of a mode-independent CQLF is restrictive and in general only holds in special cases (e.g., commuting A_r); see the surveys on common and multiple Lyapunov functions for switched linear systems [63–66]. For nonlinear switched plants, typical approaches rely on multiple Lyapunov functions and dwell-time or constrained-switching conditions [67–73]. In contrast, Koopman lifting yields a single quadratic Lyapunov function via LMIs and embeds it into the convex S-KMPC design. Mode-dependent or dwell-time Koopman formulations are left for future work [25].

3.3. Model mismatch and robust extensions

Theorems 1 (recursive feasibility) and **2** (asymptotic stability) were derived under the, in general, restrictive, but common invariance **Assumption 2** that enabling an exact Koopman lifting as proposed by [44], i.e. in the ideal Koopman setting, where the lifted predictor coincides with the true dynamics.

In practice, the EDMDC model only approximates the switched nonlinear dynamics (5) such that the following assumption imposing a proportional error bound on the one-step mismatch in lifted coordinates is more realistic, see, e.g., [74], where it was – to the best of the authors’ knowledge – first assumed for a linear surrogate model. The respective rigorous analysis can be found in the follow-up paper [75].

Assumption 6 (Proportional Error Bound). For each mode $r \in \mathcal{R}$ and all $(x, u) \in \mathbb{X} \times \mathbb{U}$, the true successor in lifted coordinates can be written as

$$\Psi(F_r(x, u)) = \mathcal{K}_r \Psi(x) + B_r u + e_r(x, u),$$

where the error term $e_r(x, u)$ satisfies

$$\|e_r(x, u)\| \leq L_x \|x\| + L_u \|u\|$$

for some constants $L_x, L_u > 0$ and all $(x, u) \in \mathbb{X} \times \mathbb{U}$. In particular, along the S-KMPC closed loop the lifted state of the real system evolves as

$$x_{\text{lift}}(k+1) = A_{\sigma(k)} x_{\text{lift}}(k) + w_k,$$

with $A_r = \mathcal{K}_r + B_r K_r^f$ and $w_k := e_{\sigma(k)}(x_k, u_k)$ bounded on $\hat{\mathbb{X}}$.

In conclusion, if only **Assumption 6** (instead of **Assumption 2**) holds, one has to extend the presented error analysis to derive the respective counterparts of **Theorems 1** and **2** analogously to [62,76,77]. In particular, we highlight the constraint tightening in [76], where state constraints are explicitly taken into account.

However, in contrast to more general Koopman surrogate models, see, e.g., the recently proposed framework [78], the verification of **Assumption 6** for linear representations remains an open question.

4. Numerical examples

We illustrate S-KMPC on two benchmarks and compare it with a *single-mode* EDMDC predictor fitted to the full data set (“classic KMPC”).

² We point out that an approximation error may render the respective closed loop only practically asymptotically stable, see, e.g. [62].

Table 1

Comparison of Predictors: Average RMSE (22) over 200 randomly sampled initial conditions.

Switching sequence 1	Average RMSE
Classic Koopman predictor	115.74%
Switched Koopman predictor	20.87%
Switching sequence 2	Average RMSE
Classic Koopman predictor	155.49%
Switched Koopman predictor	21.58%
Basis functions construction via MLP	MRMSE 1.65%

4.1. Switched system prediction

The first example is a forced van-der-Pol oscillator with two modes:

$$\begin{aligned} \dot{x}_1 &= \gamma_r x_1, \\ \dot{x}_2 &= 2x_2 - \alpha_r x_1^2 x_2 - \beta_r x_1 + u, \quad r \in \{1, 2\}, \end{aligned} \quad (21)$$

with

$$(\gamma_1, \alpha_1, \beta_1) = (2, 10, 0.8), \quad (\gamma_2, \alpha_2, \beta_2) = (3, 1, 1).$$

States and inputs are bounded by $x \in [-2, 2]^2$, $u \in [-1, 1]$. Discrete map (RK4, $T_s = 0.01$ s):

$$x_{k+1} = F_{\sigma(k)}(x_k, u_k) = \text{RK4}_{T_s}(f_{\sigma(k)})(x_k, u_k).$$

Two six-step sequences used in tests:

$$\sigma_1 = \{1, 1, 2, 2, 2, 2, 1, 1, 2, 2, 1, 1\},$$

$$\sigma_2 = \{1, 2, 1, 2, 1, 2, 1, 1, 2, 2, 1, 2\}.$$

Data generation and lifting. The ODE is discretized with $T_s = 0.01$ s. All ODEs are integrated with a fourth-order Runge–Kutta step $T_s = 0.01$ s, as in [27]. From 200 trajectories of length 1000, driven by i.i.d. $u_k \sim \mathcal{U}[-1, 1]$, we collect snapshot triples (x_k, u_k, x_{k+1}) . Observables are $M = 100$ thin-plate splines $\psi_j(x) = \|x - c_j\|^2 \log \|x - c_j\|$ with centers $c_j \sim \mathcal{U}([-1, 1]^2)$. Matrices $\mathcal{K}_r, B_r, C$ follow from the least-squares problems (10)–(11).

Results. Fig. 2 (sequence 1) and Fig. 3 (sequence 2) juxtapose the classic and proposed predictors. In both cases the S-KMPC track the true trajectories almost exactly, whereas the single-mode model drifts quickly. Table 1 reports RMSE over 200 random initial conditions: the switched predictor is an order of magnitude more accurate. A neural dictionary learned with the MLP further reduces the error to 1.65%. It was optimized with the gradient descent of the first order. For stabilization, batch normalization, see [79], was also applied. For more detail, we refer to [12]. Also, for further details pertaining to the observation of Koopman operators, please refer to [80,81].

$$\text{RMSE} = 100 \cdot \frac{\sqrt{\sum_d \|x_{\text{pred}}(dT_s) - x_{\text{true}}(dT_s)\|_2^2}}{\sqrt{\sum_d \|x_{\text{true}}(dT_s)\|_2^2}} \quad (22)$$

4.2. S-KMPC of a switched motor

The second benchmark is the armature-controlled DC motor

$$\begin{aligned} \dot{x}_1 &= -\frac{R_a^{(r)}}{L_a^{(r)}} x_1 - \frac{k_m^{(r)}}{L_a^{(r)}} x_2 u + \frac{u_a^{(r)}}{L_a^{(r)}}, \\ \dot{x}_2 &= -\frac{B^{(r)}}{J^{(r)}} x_2 + \frac{k_m^{(r)}}{J^{(r)}} x_1 u - \frac{\tau_l^{(r)}}{J^{(r)}}, \\ y &= x_2, \quad r \in \{1, 2\}, \end{aligned} \quad (23)$$

with parameters

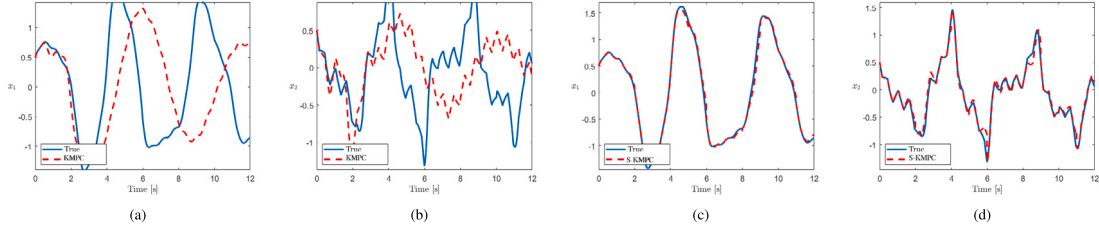


Fig. 2. The predictors compared are: (a)(b) Classic KMPC predictor with states x_1 and x_2 based on switching sequence 1 [1, 1, 2, 2, 2, 2, 1, 1, 2, 2, 1, 1] (c)(d) Proposed predictor with states x_1 and x_2 based on the same switching sequence.

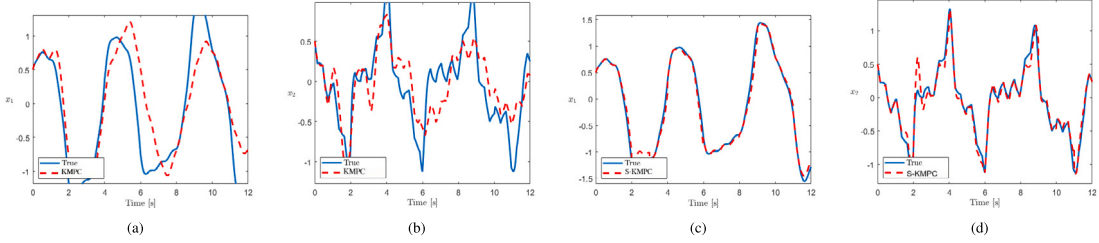


Fig. 3. The predictors compared are: (a)(b) Classic KMPC predictor with states x_1 and x_2 based on switching sequence 2 [1, 2, 1, 2, 1, 2, 1, 1, 2, 2, 1, 2] (c)(d) Proposed predictor with states x_1 and x_2 based on the same switching sequence. Both construct the basis functions with thin plate spline kernel.

$$(R_a^{(1)}, L_a^{(1)}, J^{(1)}, k_m^{(1)}, B^{(1)}, u_a^{(1)}, \tau_l^{(1)}) = (12.345, 0.314, 0.00441, 0.253, 0.00732, 60, 1.47),$$

$$(R_a^{(2)}, L_a^{(2)}, J^{(2)}, k_m^{(2)}, B^{(2)}, u_a^{(2)}, \tau_l^{(2)}) = (13.673, 0.309, 0.00413, 0.237, 0.00272, 50, 0.96).$$

Input bounds $u \in [-4, 4]$ are scaled to $[-1, 1]$, and the output is constrained to $y \in [-0.4, 0.4]$. We reuse the switching sequences σ_1, σ_2 from Section 4.1.

The rotor current is represented by x_1 , the angular velocity by x_2 and the stator current is represented by the control input u and the angular velocity is represented by the output y (a DC motor see [82]). The parameters of mode 1 are selected in accordance with the numerical setup in [27]. And for mode 2, we slightly adjust the values of certain motor parameters such as the armature voltage u_a , to simulate an alternative system mode.

Data and lifting. The ODE is discretized with $T_s = 0.01s$. From 200 random input–output trajectories ($u_k \sim \mathcal{U}[-1, 1]$, $x_0 \sim \mathcal{U}([-1, 1]^2)$) we form delay vectors with $d_e = 2$ and identify the predictors using the same $M = 100$ thin–plate splines as before (centres uniformly drawn in $[-1, 1]^3$). A neural dictionary (MLP) is trained once for comparison.

Prediction quality. Table 2 reports the RMSE and the normalized MRMSE. The switched predictor halves the error of the single-mode baseline; the MLP basis cuts it further to 6.3%.

Closed-loop test. We embed the predictors in MPC with horizon N_h , weights $Q = Q_{N_h} = 1$, $R = 10^{-2}$, and track (i) a step, (ii) $y_r(t) = 0.5 \cos(2\pi t/3)$. Figs. 4–5 show that S-KMPC respects all constraints and matches the classic KMPC tracking error while reducing on-line solve time from 21.8s to 10.1s per step. Training the MLP basis takes 209s offline but yields the most accurate predictor.

$$\text{MRMSE} = 100 \cdot \frac{\sqrt{\frac{1}{d} \sum_d \|x_{\text{pred}}(dT_s) - x_{\text{true}}(dT_s)\|_2^2}}{\|\max(x_{\text{true}}) - \min(x_{\text{true}})\|_2} \quad (24)$$

Overall, S-KMPC combines a prediction-error reduction with a speed-up in control computation—while retaining a purely data-driven workflow.

Remark 8. Prediction–measurement gaps in S-KMPC chiefly arise from: (i) representation limits—fixed observables may miss fast transients, saturation, or sharp state-dependent couplings; (ii) finite-data

Table 2

Average prediction errors over 200 random initializations.

Predictor	MRMSE [%]	RMSE [%]
Single-mode (seq. 1)	58.98	118.0
Single-mode (seq. 2)	57.98	116.0
Switched (seq. 1)	33.58	85.2
Switched (seq. 2)	22.60	55.8
Switched + MLP (seq. 2)	6.31	–

identification—per-mode Koopman matrices incur sampling error decaying with the number of snapshots [32]; (iii) unmodeled/time-varying effects—e.g., friction, load shifts, or cross-mode coupling; these can be handled via online learning of an additive linear disturbance [83]. Identifying the dominant source suggests, respectively: enlarge the basis, gather more mode-specific data, or add a residual term to improve closed-loop accuracy.

Remark 9. The benchmarks are low-order and noise-free to isolate lifting effects. S-KMPC is dimension-agnostic and compatible with robust tightening and disturbance-augmented predictors [84]. Richer predictors for switched systems – bilinear Koopman with generator interpolation [25], neural hybrid models [24], and Monte-Carlo optimization [10] – may boost accuracy but forfeit the fully linear lifted structure enabling (i) a CQLF via one LMI and (ii) a small convex QP online. Achieving the same mode-uniform stability guarantees and real-time efficiency with bilinear or neural surrogates remains open.

4.3. Switching sequence optimization

To illustrate switching-sequence optimization, we assess S-KMPC on a two-mode Duffing oscillator with mode-dependent coefficients

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -a_r x_1 - b_r x_2 - \beta x_1^3 + u, \quad r \in \{1, 2\}, \end{aligned} \quad (25)$$

with parameters

$$\beta = 1.0, \quad (a_1, b_1) = (1.1, 0.8), \quad (a_2, b_2) = (0.5, 0.35).$$

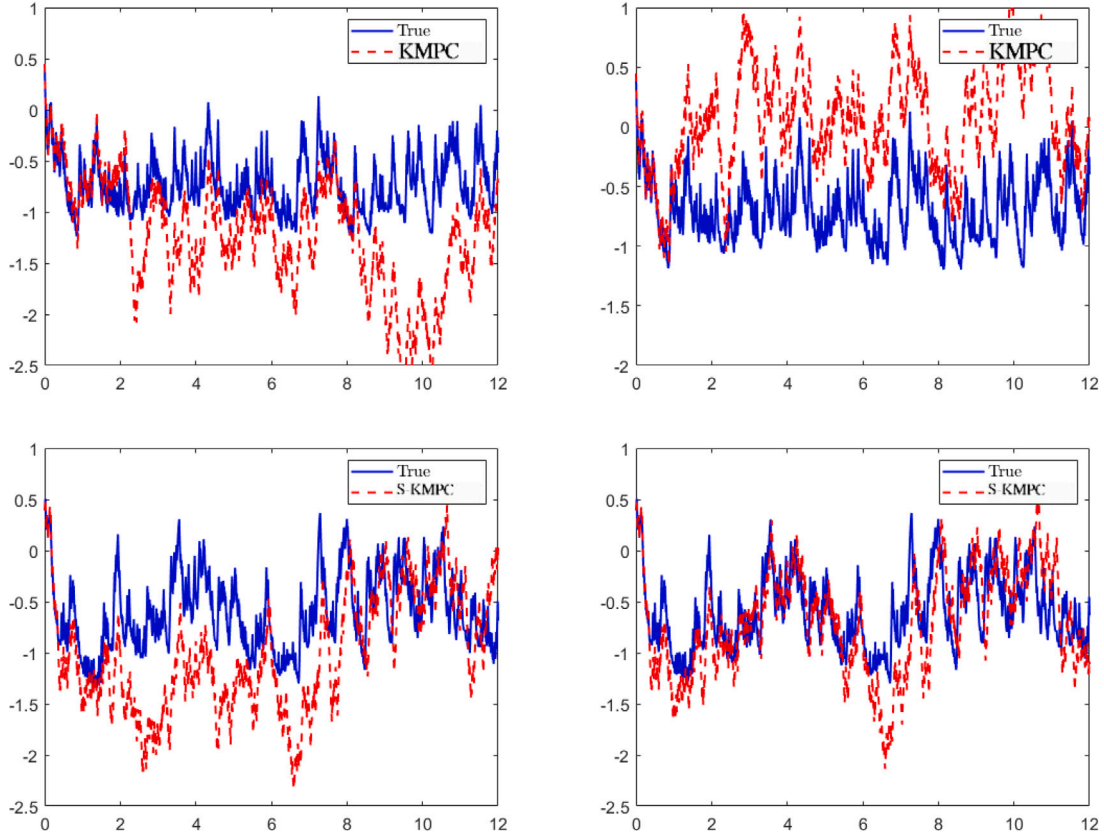


Fig. 4. The first two are classic predictor with outputs based on switching sequence 1 and 2 respectively. And the last two give the prediction of the S-KMPC controller under switching sequence 1 and 2 respectively.

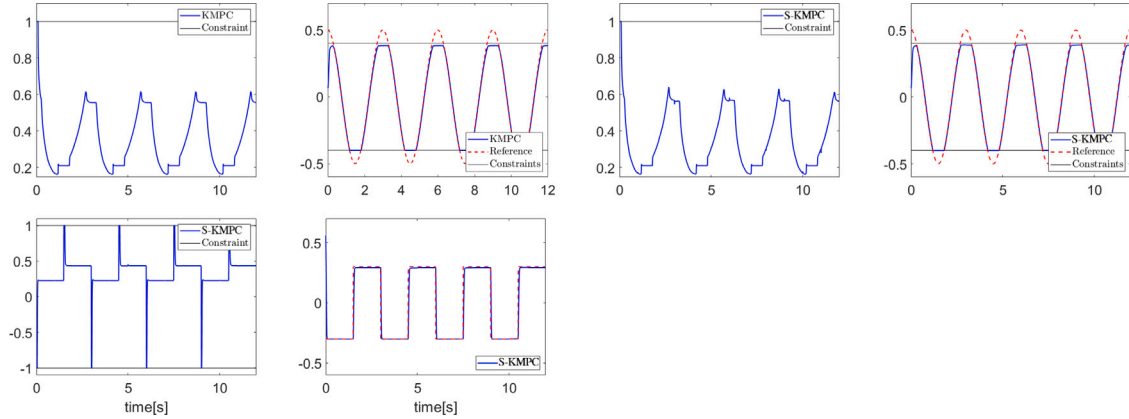


Fig. 5. The first two subplots depict the control signal and output of classic MPC controller based on switching sequence 1. The following two sub-pictures depict control signals and output tracking when switching sequence 2 is applied, with reference to $y_r(t) = 0.5 \cos(2\pi t/3)$, while the last two refer to the step signal.

For each mode $r \in \{1, 2\}$, we identify a lifted EDMDc predictor using a thin-plate-spline dictionary and a common decoder C . In lifted coordinates, the measured output is $y = C_j C x_{\text{lift}}$, and one-step prediction errors are on the order of 10^{-6} ; cf. Fig. 7.

At each sampling instant k (sampling time $\Delta t = 0.01$ s), we generate a small library S_k of admissible short mode prefixes, anchored at the currently active mode. For each candidate prefix $\sigma \in S_k$, we: (i) complete the prediction horizon by holding the last prefix mode, and (ii) solve the convex S-KMPC QP in the lifted coordinates, with quadratic output-tracking and input penalties, subject to the lifted dynamics, input bounds and decoded state constraints. The resulting

closed-loop cost $J_{\text{QP}}(\sigma)$ is augmented by a simple switching penalty proportional to the number of mode changes in the prefix. Among all candidates, we select the prefix that minimizes this penalized cost and apply its first mode and input.

Fig. 6 shows that the controller achieves accurate tracking of a time-varying reference $y_{\text{ref}}(t)$ with bounded inputs, while the chosen mode sequence in Fig. 6(b) remains sparse: switches occur only when they improve the overall objective. The one-step prediction in Koopman space stays essentially exact, see Fig. 7, confirming that the data-driven lifted model can support on-line schedule selection without destroying the convex QP structure.

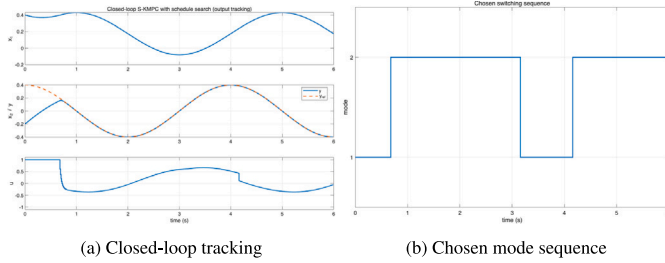


Fig. 6. Closed-loop tracking.

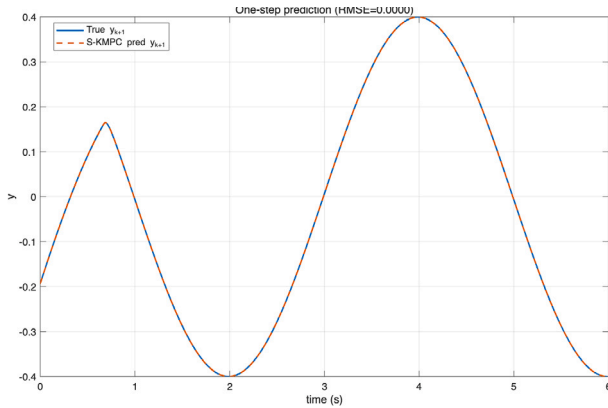


Fig. 7. One-step S-KMPC prediction vs. truth (output).

We emphasize that this example follows the S-KMPC structure of Section 3.1 and demonstrates the practicality of schedule optimization, but we do not attempt a formal stability proof for this time-varying tracking setup with switching penalties; the theoretical results in Section 3.2 are stated for the nominal regulation case.

5. Conclusion and future work

In this paper, we proposed a S-KMPC approach for discrete-time switched nonlinear systems. By leveraging Koopman theory and ED-MDC, we lifted the nonlinear switched dynamics into a linear representation, obtaining a data-driven linear predictor that can be embedded into a convex MPC scheme. With appropriate terminal constraints and feasibility assumptions, the proposed S-KMPC scheme ensures both mode-uniform recursive feasibility and closed-loop stability in an ideal Koopman setting. Numerical simulations on representative benchmarks demonstrated accurate prediction, good closed-loop performance, and competitive computational effort. Future work will incorporate explicit model-mismatch bounds via robust constraint tightening and extend the analysis to stochastic switching, further broadening the applicability of Koopman-based control.

CRediT authorship contribution statement

Xuqian Ju: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Qing Sun:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Dario Dennstädt:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Karl Worthmann:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Dajun Du:** Resources, Funding acquisition. **Minrui Fei:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] J. Chen, C. Dou, L. Xiao, Z. Wang, Fusion state estimation for power systems under DoS attacks: A switched system approach, *IEEE Trans. Syst. Man Cybern.: Syst.* 49 (8) (2019) 1679–1687, <http://dx.doi.org/10.1109/TSMC.2019.2895912>.
- [2] H. Saranathan, M.J. Grant, Relaxed autonomously switched hybrid system approach to indirect multiphase aerospace trajectory optimization, *J. Spacecr. Rockets* 55 (3) (2018) 611–621, <http://dx.doi.org/10.2514/1.A34012>.
- [3] A. Anderson, A. González, A. Ferramosca, E. Hernandez-Vargas, Discrete-time MPC for switched systems with applications to biomedical problems, *Commun. Nonlinear Sci. Numer. Simul.* 95 (2021) 105586, <http://dx.doi.org/10.1016/j.cnsns.2020.105586>.
- [4] J. Jin, Y. Kim, S. Wee, D. Lee, N. Gans, A stable switched-system approach to collision-free wheeled mobile robot navigation, *J. Intell. Robot. Syst.* 86 (2017) 599–616, <http://dx.doi.org/10.1007/s10846-017-0467-z>.
- [5] N. Xu, Y. Chen, A. Xue, H. Wang, X. Zhao, Backstepping-based controller design for uncertain switched high-order nonlinear systems via PI compensation, *IEEE Trans. Syst. Man Cybern.: Syst.* 52 (12) (2022) 7810–7820, <http://dx.doi.org/10.1109/TSMC.2022.3164893>.
- [6] D. Cui, Y. Wu, Z. Xiang, Finite-time adaptive fault-tolerant tracking control for nonlinear switched systems with dynamic uncertainties, *Internat. J. Robust Nonlinear Control* 31 (8) (2021) 2976–2992, <http://dx.doi.org/10.1002/rnc.5429>.
- [7] J. Daafouz, P. Riedinger, C. Iung, Stability analysis and control synthesis for switched systems: a switched Lyapunov function approach, *IEEE Trans. Autom. Control* 47 (11) (2002) 1883–1887, <http://dx.doi.org/10.1109/TAC.2002.804474>.
- [8] S. Li, C.K. Ahn, J. Guo, Z. Xiang, Neural network-based sampled-data control for switched uncertain nonlinear systems, *IEEE Trans. Syst. Man Cybern.: Syst.* 51 (9) (2021) 5437–5445, <http://dx.doi.org/10.1109/TSMC.2019.2954231>.
- [9] Y. Lan, I. Mezić, Linearization in the large of nonlinear systems and Koopman operator spectrum, *Phys. D: Nonlinear Phenom.* 242 (1) (2013) 42–53, <http://dx.doi.org/10.1016/j.physd.2012.08.017>.
- [10] J. Hespanha, K. Çamsarı, Markov chain Monte Carlo for Koopman-based optimal control, *IEEE Control. Syst. Lett.* 8 (2024) 1901–1906, <http://dx.doi.org/10.1109/LCSYS.2024.3411930>.
- [11] R. Strässer, K. Worthmann, I. Mezić, J. Berberich, M. Schaller, F. Allgöwer, An overview of Koopman-based control: From error bounds to closed-loop guarantees, *Annu. Rev. Control.* 61 (2026) 101035, <http://dx.doi.org/10.1016/j.arcontrol.2025.101035>.
- [12] N. Takeishi, Y. Kawahara, T. Yairi, Learning Koopman invariant subspaces for dynamic mode decomposition, in: I. Guyon, U.V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), in: *Advances in Neural Information Processing Systems*, vol. 30, Curran Associates, Inc. 2017, pp. 1130–1140, URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3a835d3215755c435ef4fe9965a3f2a0-Paper.pdf.
- [13] M. Haseli, J. Cortés, Recursive forward-backward EDMD: Guaranteed algebraic search for Koopman invariant subspaces, *IEEE Access* 13 (2025) 61006–61025, <http://dx.doi.org/10.1109/ACCESS.2025.3554154>.
- [14] S.L. Brunton, B.W. Brunton, J.L. Proctor, J.N. Kutz, Koopman invariant subspaces and finite linear representations of nonlinear dynamical systems for control, *PLoS One* 11 (2) (2016) 1–19, <http://dx.doi.org/10.1371/journal.pone.0150171>.
- [15] M. Korda, I. Mezić, Optimal construction of Koopman eigenfunctions for prediction and control, *IEEE Trans. Autom. Control* 65 (12) (2020) 5114–5129, <http://dx.doi.org/10.1109/TAC.2020.2978039>.
- [16] A. Mauroy, J. Goncalves, Koopman-based lifting techniques for nonlinear systems identification, *IEEE Trans. Autom. Control* 65 (6) (2020) 2550–2565, <http://dx.doi.org/10.1109/TAC.2019.2941433>.
- [17] S.L. Brunton, M. Budišić, E. Kaiser, J.N. Kutz, Modern Koopman theory for dynamical systems, *SIAM Rev.* 64 (2) (2022) 229–340, <http://dx.doi.org/10.1137/21M1401243>.
- [18] M.O. Williams, I.G. Kevrekidis, C.W. Rowley, A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition, *J. Nonlinear Sci.* 25 (6) (2015) 1307–1346, <http://dx.doi.org/10.1007/s00332-015-9258-5>.
- [19] F. Köhne, F.M. Philipp, M. Schaller, A. Schiela, K. Worthmann, L^∞ -error bounds for approximations of the Koopman operator by kernel extended dynamic mode decomposition, *SIAM J. Appl. Dyn. Syst.* 24 (1) (2025) 501–529, <http://dx.doi.org/10.1137/24M1650120>.

- [20] M. Laurini, L. Consolini, M. Locatelli, Fast numerical solution of optimal control problems for switched systems: An application to path planning, in: 2020 59th IEEE Conference on Decision and Control, CDC, 2020, pp. 4105–4110, <http://dx.doi.org/10.1109/CDC42340.2020.9304280>.
- [21] S. Peitz, S. Klus, Koopman operator-based model reduction for switched-system control of PDEs, *Automatica* 106 (2019) 184–191, <http://dx.doi.org/10.1016/j.automatica.2019.05.016>.
- [22] S.H. Son, H.-K. Choi, J. Moon, J.S.-I. Kwon, Hybrid Koopman model predictive control of nonlinear systems using multiple EDMD models: An application to a batch pulp digester with feed fluctuation, *Control Eng. Pract.* 118 (2022) 104956, <http://dx.doi.org/10.1016/j.conengprac.2021.104956>.
- [23] Z. Ning, X. Fang, Y. Li, L. Xie, Learning-based predictive control of switched nonlinear systems via Koopman modeling approach, *J. Syst. Sci. Complex.* 38 (2) (2025) 691–716, <http://dx.doi.org/10.1007/s11424-025-4546-0>.
- [24] C. Bakker, A. Bhattacharya, S. Chatterjee, C.J. Perkins, M.R. Oster, Learning Koopman representations for hybrid systems, 2020, [arXiv:2006.12427](https://arxiv.org/abs/2006.12427).
- [25] S. Peitz, S.E. Otto, C.W. Rowley, Data-driven model predictive control using interpolated Koopman generators, *SIAM J. Appl. Dyn. Syst.* 19 (3) (2020) 2162–2193, <http://dx.doi.org/10.1137/20m1325678>.
- [26] J.L. Proctor, S.L. Brunton, J.N. Kutz, Generalizing Koopman theory to allow for inputs and control, *SIAM J. Appl. Dyn. Syst.* 17 (1) (2018) 909–930, <http://dx.doi.org/10.1137/16M1062296>.
- [27] M. Korda, I. Mezić, Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control, *Automatica* 93 (2018) 149–160, <http://dx.doi.org/10.1016/j.automatica.2018.03.046>.
- [28] M. Budišić, R. Mohr, I. Mezić, Applied Koopmanism, *Chaos: An Interdiscip. J. Nonlinear Sci.* 22 (4) (2012) 047510, <http://dx.doi.org/10.1063/1.4772195>.
- [29] A. Mauroy, I. Mezić, Y. Susuki, The Koopman operator in systems and control, first ed., in: *Lecture Notes in Control and Information Sciences*, vol. 484, Springer Nature, 2020, <http://dx.doi.org/10.1007/978-3-030-35713-9>.
- [30] A. Mauroy, Y. Susuki, I. Mezić, Introduction to the Koopman operator in dynamical systems and control theory, in: *The Koopman Operator in Systems and Control: Concepts, Methodologies, and Applications*, Springer, 2016, pp. 3–49, http://dx.doi.org/10.1007/978-3-319-48764-7_1.
- [31] L.C. Jacob, R. Tóth, M. Schoukens, Koopman form of nonlinear systems with inputs, *Automatica* 162 (2024) 111525, <http://dx.doi.org/10.1016/j.automatica.2024.111525>.
- [32] F. Nüske, S. Peitz, F. Philipp, M. Schaller, K. Worthmann, Finite-data error bounds for Koopman-based prediction and control, *J. Nonlinear Sci.* 33 (2023) 14, <http://dx.doi.org/10.1007/s00332-022-09862-1>.
- [33] M. Schaller, K. Worthmann, F.M. Philipp, S. Peitz, F. Nüske, Towards reliable data-based optimal and predictive control using extended DMD, *IFAC-PapersOnLine* 56 (1) (2023) 169–174, <http://dx.doi.org/10.1016/j.ifacol.2023.02.029>.
- [34] L. Bold, L. Grüne, M. Schaller, K. Worthmann, Data-driven MPC with stability guarantees using extended dynamic mode decomposition, *IEEE Trans. Autom. Control* 70 (1) (2025) 534–541, <http://dx.doi.org/10.1109/TAC.2024.3431169>.
- [35] R. Strässer, M. Schaller, K. Worthmann, J. Berberich, F. Allgöwer, Koopman-based feedback design with stability guarantees, *IEEE Trans. Autom. Control* 70 (1) (2025) 355–370, <http://dx.doi.org/10.1109/TAC.2024.3425770>.
- [36] L. Bold, F.M. Philipp, M. Schaller, K. Worthmann, Kernel-based Koopman approximants for control: Flexible sampling, error analysis, and stability, *SIAM J. Control Optim.* 63 (2025) 3825–4114, <http://dx.doi.org/10.1137/24M1715945>.
- [37] H. Wendland, *Scattered data approximation*, in: *Cambridge Monographs on Applied and Computational Mathematics*, Cambridge University Press, 2004, <http://dx.doi.org/10.1017/CBO9780511617539>.
- [38] D. Liberzon, *Switching in systems and control*, first ed., in: *Communications and Control Engineering*, vol. 190, Birkhäuser Boston, Boston, 2003, URL <https://link.springer.com/book/10.1007/978-1-4612-0017-8>.
- [39] C. Mugisho Zagabe, A. Mauroy, Switched nonlinear systems in the Koopman operator framework: Toward a Lie-algebraic condition for uniform stability, in: 2021 European Control Conference, ECC, 2021, pp. 281–286, <http://dx.doi.org/10.23919/ECC54610.2021.9655200>.
- [40] J. Uralde, O. Barambones, A. del Rio, I. Calvo, E. Artetxe, Rule-based operation mode control strategy for the energy management of a fuel cell electric vehicle, *Batteries* 10 (6) (2024) 214, <http://dx.doi.org/10.3390/batteries10060214>.
- [41] X. Wang, B. Lv, K. Wang, R. Zhang, Autonomous switching of task-level strategies, *Int. J. Appl. Math. Comput. Sci.* 33 (4) (2023) 553–568, <http://dx.doi.org/10.34768/amcs-2023-0040>.
- [42] D. Goswami, D.A. Paley, Global bilinearization and controllability of control-affine nonlinear systems: A Koopman spectral approach, in: *IEEE 56th Annual Conference on Decision and Control, CDC, 2017*, pp. 6107–6112, <http://dx.doi.org/10.1109/CDC.2017.8264582>.
- [43] D. Goswami, D.A. Paley, Bilinearization, reachability, and optimal control of control-affine nonlinear systems: A Koopman spectral approach, *IEEE Trans. Autom. Control* 67 (6) (2021) 2715–2728, <http://dx.doi.org/10.1109/TAC.2021.3088802>.
- [44] X. Shang, J. Cortés, Y. Zheng, Willems' fundamental lemma for nonlinear systems with Koopman linear embedding, *IEEE Control. Syst. Lett.* 8 (2024) 3135–3140, <http://dx.doi.org/10.1109/LCSYS.2024.3522594>.
- [45] X. Shang, J. Cortés, Y. Zheng, On the exponential stability of Koopman model predictive control, 2025, [arXiv:2511.02008](https://arxiv.org/abs/2511.02008).
- [46] M.C. Honecker, H. Gernandt, K. Wulff, C. Trunk, J. Reger, Feedback rectifiable pairs and stabilization of switched linear systems, *Systems Control Lett.* 186 (2024) 105755, <http://dx.doi.org/10.1016/j.sysconle.2024.105755>.
- [47] M.O. Williams, C.W. Rowley, I. Mezić, I.G. Kevrekidis, Data fusion via intrinsic dynamic variables: An application of data-driven Koopman spectral analysis, *Europhys. Lett.* 109 (4) (2015) 40007, <http://dx.doi.org/10.1209/0295-5075/109/40007>.
- [48] X. Huang, W. Zhou, X. Li, L. Zhu, Online Koopman operator learning to identify cross-coupling effect of piezoelectric tube scanners in atomic force microscopes, *IEEE Trans. Ind. Inform.* 17 (11) (2021) 7601–7611, <http://dx.doi.org/10.1109/TII.2021.3081418>.
- [49] S. Sinha, S.P. Nandanoori, D.A. Barajas-Solano, Online real-time learning of dynamical systems from noisy streaming data, *Sci. Rep.* 13 (2023) 22564, <http://dx.doi.org/10.1038/s41598-023-49045-w>.
- [50] X. Dong, Y.-L. Bai, Y. Lu, M. Fan, An improved sparse identification of nonlinear dynamics with Akaike information criterion and group sparsity, *Nonlinear Dynam.* 111 (3) (2023) 1485–1510, <http://dx.doi.org/10.1007/s11071-022-07875-9>.
- [51] M.O. Williams, C.W. Rowley, I.G. Kevrekidis, A kernel-based method for data-driven Koopman spectral analysis, *J. Comput. Dyn.* 2 (2) (2015) 247–265, <http://dx.doi.org/10.3934/jcd.2015005>.
- [52] F. Takens, Detecting strange attractors in turbulence, in: *Dynamical Systems and Turbulence*, Warwick 1980, in: *Lecture Notes in Mathematics*, vol. 898, Springer Berlin Heidelberg, Berlin, Heidelberg, 2006, pp. 366–381, <http://dx.doi.org/10.1007/BFb0091924>.
- [53] A. Ziesler, M. Dellnitz, R. Gerlach, The numerical computation of unstable manifolds for infinite dimensional dynamical systems by embedding techniques, *SIAM J. Appl. Dyn. Syst.* 18 (3) (2019) 1265–1292, <http://dx.doi.org/10.1137/18m1204395>.
- [54] J. Singh, R. Banerjee, A study on single and multi-layer perceptron neural network, in: 2019 3rd International Conference on Computing Methodologies and Communication, ICCMC, 2019, pp. 35–40, <http://dx.doi.org/10.1109/ICCMC.2019.8819775>.
- [55] K. He, X. Zhang, S. Ren, J. Sun, Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification, in: 2015 IEEE International Conference on Computer Vision, ICCV, IEEE Computer Society, Los Alamitos, CA, USA, 2015, pp. 1026–1034, <http://dx.doi.org/10.1109/ICCV.2015.123>.
- [56] M. Blischke, J.P. Hespanha, Learning switched Koopman models for control of entity-based systems, in: 2023 62nd IEEE Conference on Decision and Control, CDC, 2023, pp. 6006–6013, <http://dx.doi.org/10.1109/CDC49753.2023.10384106>.
- [57] L. Grüne, J. Pannek, Nonlinear model predictive control, in: *Nonlinear Model Predictive Control: Theory and Algorithms*, Springer International Publishing, Cham, 2017, pp. 45–69, http://dx.doi.org/10.1007/978-3-319-46024-6_3.
- [58] O. Mason, R. Shorten, On common quadratic Lyapunov functions for stable discrete-time LTI systems, *IMA J. Appl. Math.* 69 (3) (2004) 271–283, <http://dx.doi.org/10.1093/imamat/69.3.271>.
- [59] M.M. Moldovan, M.S. Gowda, On common linear/quadratic Lyapunov functions for switched linear systems, in: *Nonlinear Analysis and Variational Problems: In Honor of George Isac*, Springer New York, New York, NY, 2010, pp. 415–429, http://dx.doi.org/10.1007/978-1-4419-0158-3_25.
- [60] M.D. Rossa, Converse Lyapunov results for switched systems with lower and upper bounds on switching intervals, *Automatica* 163 (2024) 111576, <http://dx.doi.org/10.1016/j.automatica.2024.111576>.
- [61] M.D. Rossa, A. Tanwani, L. Zaccarian, Max–min Lyapunov functions for switched systems and related differential inclusions, *Automatica* 120 (2020) 109123, <http://dx.doi.org/10.1016/j.automatica.2020.109123>.
- [62] K. Worthmann, R. Strässer, M. Schaller, J. Berberich, F. Allgöwer, Data-driven MPC with terminal conditions in the Koopman framework, in: 63rd IEEE Conference on Decision and Control, CDC, 2024, pp. 146–151, <http://dx.doi.org/10.1109/cdc56724.2024.10886773>.
- [63] D. Liberzon, R. Tempo, Common Lyapunov functions and gradient algorithms, *IEEE Trans. Autom. Control* 49 (6) (2004) 990–994, <http://dx.doi.org/10.1109/TAC.2004.829632>.
- [64] H. Lin, P.J. Antsaklis, Stability and stabilizability of switched linear systems: a survey of recent results, *IEEE Trans. Autom. Control* 54 (2) (2009) 308–322, <http://dx.doi.org/10.1109/TAC.2008.2012009>.
- [65] R. Shorten, F. Wirth, O. Mason, K. Wulff, C. King, Stability criteria for switched and hybrid systems, *SIAM Rev.* 49 (4) (2007) 545–592, <http://dx.doi.org/10.1137/05063516X>.
- [66] X. Zhao, L. Zhang, P. Shi, M. Liu, Stability and stabilization of switched linear systems with mode-dependent average dwell time, *IEEE Trans. Autom. Control* 57 (7) (2011) 1809–1815, <http://dx.doi.org/10.1109/TAC.2011.2178629>.
- [67] X. Zhao, X. Zheng, B. Niu, L. Liu, Adaptive tracking control for a class of uncertain switched nonlinear systems, *Automatica* 52 (2015) 185–191, <http://dx.doi.org/10.1016/j.automatica.2014.11.019>.

- [68] M.J. Mirzaei, E. Aslmostafa, M. Asadollahi, M.A. Badamchizadeh, Robust adaptive finite-time stabilization control for a class of nonlinear switched systems based on finite-time disturbance observer, *J. Franklin Inst.* 358 (7) (2021) 3332–3352, <http://dx.doi.org/10.1016/j.jfranklin.2021.02.010>.
- [69] J. Fu, R. Ma, T. Chai, Global finite-time stabilization of a class of switched nonlinear systems with the powers of positive odd rational numbers, *Automatica* 54 (2015) 360–373, <http://dx.doi.org/10.1016/j.automatica.2015.02.023>.
- [70] R. Wang, L. Hou, G. Zong, S. Fei, D. Yang, Stability and stabilization of continuous-time switched systems: a multiple discontinuous convex Lyapunov function approach, *Internat. J. Robust Nonlinear Control* 29 (5) (2019) 1499–1514, <http://dx.doi.org/10.1002/rnc.4449>.
- [71] X. Lin, X. Li, J.H. Park, Output-feedback stabilization for planar output-constrained switched nonlinear systems, *Internat. J. Robust Nonlinear Control* 30 (5) (2020) 1819–1830, <http://dx.doi.org/10.1002/rnc.4850>.
- [72] J. Huang, J. Cheng, X. Lin, J.H. Park, Adaptive fixed-time non-smooth stabilization of switched systems with output constraints, *Internat. J. Robust Nonlinear Control* 35 (6) (2025) 1993–2009, <http://dx.doi.org/10.1002/rnc.7771>.
- [73] H. Liu, X. Li, H. Wang, Adaptive prescribed finite-time asymptotic tracking control for switched systems with unknown initial conditions and full-state constraints, *Int. J. Syst. Sci.* 55 (2) (2024) 370–390, <http://dx.doi.org/10.1080/002071721.2023.2272219>.
- [74] R. Strässer, J. Berberich, F. Allgöwer, Robust data-driven control for nonlinear systems using the Koopman operator, *IFAC-PapersOnLine* 56 (2) (2023) 2257–2262, <http://dx.doi.org/10.1016/j.ifacol.2023.10.1190>.
- [75] R. Strässer, M. Schaller, K. Worthmann, J. Berberich, F. Allgöwer, SafEDMD: A Koopman-based data-driven controller design framework for nonlinear dynamical systems, *Automatica* 185 (2025) 112732, <http://dx.doi.org/10.1016/j.automatica.2025.112732>.
- [76] I. Schimperna, L. Bold, J. Köhler, K. Worthmann, L. Magni, Stability of data-driven Koopman MPC with terminal conditions, 2025, arXiv preprint [arXiv:2511.21248](https://arxiv.org/abs/2511.21248).
- [77] S.J. Kuntz, J.B. Rawlings, Beyond inherent robustness: strong stability of MPC despite plant-model mismatch, 2025, arXiv preprint [arXiv:2411.15452](https://arxiv.org/abs/2411.15452).
- [78] I. Schimperna, K. Worthmann, M. Schaller, L. Bold, L. Magni, Data-driven model predictive control: Asymptotic stability despite approximation errors exemplified in the Koopman framework, 2025, arXiv preprint [arXiv:2505.05951](https://arxiv.org/abs/2505.05951).
- [79] S. Ioffe, C. Szegedy, Batch normalization: Accelerating deep network training by reducing internal covariate shift, in: *Proceedings of the 32nd International Conference on Machine Learning*, in: *Proceedings of Machine Learning Research*, vol. 37, PMLR, Lille, France, 2015, pp. 448–456, URL <https://arxiv.org/abs/1502.03167>.
- [80] S. Otto, S. Peitz, C. Rowley, Learning bilinear models of actuated Koopman generators from partially observed trajectories, *SIAM J. Appl. Dyn. Syst.* 23 (1) (2024) 885–923, <http://dx.doi.org/10.1137/22M1523601>.
- [81] S. Peitz, H. Harder, F. Nüske, F.M. Philipp, M. Schaller, K. Worthmann, Equivariance and partial observations in Koopman operator theory for partial differential equations, *J. Comput. Dyn.* 12 (2) (2025) 305–324, <http://dx.doi.org/10.3934/jcd.2024035>.
- [82] S. Daniel-Berhe, H. Unbehauen, Experimental physical parameter estimation of a thyristor driven DC-motor using the HMF-method, *Control Eng. Pract.* 6 (5) (1998) 615–626, [http://dx.doi.org/10.1016/S0967-0661\(98\)00036-7](http://dx.doi.org/10.1016/S0967-0661(98)00036-7).
- [83] D. Bruder, D. Bombara, R.J. Wood, A Koopman-based residual modeling approach for the control of a soft robot arm, *Int. J. Robot. Res.* 44 (1) (2024) <http://dx.doi.org/10.1177/02783649241272114>.
- [84] S.H. Son, A. Narasingam, J.S.-I. Kwon, Development of offset-free Koopman Lyapunov-based model predictive control and mathematical analysis for zero steady-state offset condition considering influence of Lyapunov constraints on equilibrium point, *J. Process Control* 118 (2022) 26–36, <http://dx.doi.org/10.1016/j.jprocont.2022.08.005>.